Greedy Approx: Best Possible

From previous lectures, we have that

Greedy al. guarantees $(1-1/e)$-approx'n to:

$$\max: f(S)$$
$$\text{st}: |S| \leq k \qquad \text{(uniform matroid constraint)}$$

when f is submodular & monotone.

Thm: (Feige) $\forall \varepsilon > 0$, it is NP-hard to achieve a $(1-1/e+\varepsilon)$ approx'n for the Max-k-Cover problem.

Max-k-Cover: Given a ground set E, and subsets $S_1, S_2, S_3, \ldots, S_m \subseteq E$ and $k \leq |E|$, solve:

$$\max: \left|\bigcup_{i \in I} S_i\right|$$
$$\text{st}: |I| \leq k$$

Matroid Constraint

Instead of a uniform matroid: Let $\mathcal{I}$ be the independent sets of a general matroid:

$$\boxed{\begin{array}{l} \max: f(I) \\ \text{st}: I \in \mathcal{I}. \end{array}}$$

Ex: The Submodular Welfare Problem

n items, m agents. Each item must go to 1 agent

each agent receives b/w 0 and n items. Assume the utility of agent i is a monotone submodular function: $f_i(S)$

Goal: Allocate n items to maximize $\sum_{i=1}^{m} f_i(S_i)$

Exercise: Show that the S.W.P. is of the form

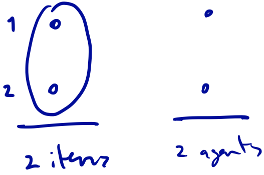
$$\max: f(I)$$

$$\text{st}: I \in \mathcal{I}, \quad \mathcal{I} \text{ a matroid}, \quad f \text{ monotone, sub.}$$

Greedy Gives 1/2-Approx

An instance of above problem:

$n = m = 2$



$$w_1(S) = \begin{cases} 0 & 1 \notin S \\ 1 & \text{if } 1 \in S \end{cases}$$

$$w_2(S) = \begin{cases} 1 & \text{if } S = \{2\} \\ 1+\varepsilon & \text{if } S = \{1\} \\ 1+\varepsilon & \text{if } S = \{1,2\} \end{cases}$$

Greedy: Gives elt 1 to agent 2.

Aggregate welfare: $w_1(S) + w_2(E \setminus S)$

Greedy: $(1+\varepsilon)$ // Greedy is $1/2$-approx'n.

Opt: 2

Greedy Always Gives 1/2-Approx

Thm: For f a monotone submodular function, and $\mathcal{I}$ any matroid, the Greedy alg gives a $\frac{1}{2}$-approx to the problem:

$$\max: f(I) \quad \text{st: } I \in \mathcal{I}.$$

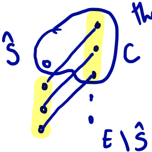
pf: $\hat{S}$ be the greedy sol'n: $\hat{S} = \{\underbrace{g_1, g_2, g_3, \ldots,}_{} g_k\}$

$S_i = \{g_1, \ldots, g_i\}$, $S_k = \hat{S}$.

Let $C = \{o_1, \ldots, o_k\}$ denote opt solution.

Note: monotonicity $\Rightarrow$ $\hat{S}$ & C bases of matroid.

Recall: We proved a strong exchange property: In the directed graph $D(\hat{S})$ there is a perfect matching on $\hat{S} \Delta C$



Thus: we can re-order elts $o_1, \ldots, o_k$ if necessary, so that

$$\hat{S} - g_i + o_i \in \mathcal{I}.$$

Proof — continued —

By def'n of Greedy:

$$f_{S_i}(g_{i+1}) \geq f_{S_i}(o_{i+1}) \quad (*)$$

$$\underline{f(\hat{S})} = \sum_{i=1}^{k-1} f(S_{i+1}) - f(S_i)$$

$$= \sum_{i=1}^{k-1} f_{S_i}(g_{i+1})$$

$$\geq \sum f_{S_i}(o_{i+1})$$

$$\geq \sum f_{\hat{S}}(o_{i+1}) \qquad \text{(submodularity)}$$

$$\geq f_{\hat{S}}(C) = \underline{\underline{f(C \cup \hat{S})} - f(\hat{S})} \qquad \text{(submodularity)}$$

$$\geq f(C) - \underline{f(\hat{S})} \qquad \text{by monotonicity.}$$

$$\Rightarrow \quad f(\hat{S}) \geq \tfrac{1}{2} f(C).$$